

## Mysterious Sum (sum)

Luca is going to Bologna by train to attend the grand final of OIS's current edition, but a recent accident blocked the high-speed railway and made the journey way longer. To kill time, Luca loves to solve some puzzles and he is now working on an interesting one.

In this kind of puzzle, you are given a simple sum of two integers numbers and the result of this addition, but there is one caveat. Operands' digits have been substituted with letters, like in the next example, following a rule commonly known as “same digit, same letter”.

$$\begin{array}{r} A \ B \ + \\ C \ A \ = \\ 5 \ 8 \end{array}$$

More precisely, this means that every digit from 0 to 9 has been linked to an uppercase letter from A to J and every occurrence has been substituted with that letter.

Moreover, the mapping always links different digits to different letters and vice versa, so that no two digits have the same associated letter and that no two letters are associated with the same digit.

Luca is still a bit puzzled and his train ride is almost over: can you recover the two addends for him?

🔗 Among the attachments of this task you may find a template file `sum.*` with a sample incomplete implementation.

## Input

The first two lines contain respectively the two addends with digits substituted with letters. Both the addends have the same length  $N$ . The third line contains the result given by the sum of the two operands.

## Output

You need to write two lines which contain respectively the digits of the first and the second addend so that their sum equals the result given in input.

## Constraints

- $1 \leq N \leq 10\,000$ .
- Addends are written as valid numbers: they *never* have leading zeros (unless that the whole number is just a 0). To exemplify, 000 and 012 are not valid numbers, while 0 and 123 are.
- It is guaranteed that a solution exists.
- If there are multiple solutions, you can output any of them.



Figure 1: “La settimana enigmistica” is a very popular Italian weekly publication. This is the first issue, published on 23rd January 1932!

# Scoring

Your program will be tested against several test cases grouped in subtasks. In order to obtain the score of a subtask, your program needs to correctly solve all of its test cases.

- Subtask 1 (0 points)      Examples.
- Subtask 2 (10 points)       $N = 1$ .
- Subtask 3 (30 points)      Summing the two operands *never* involves a carry.
- Subtask 4 (40 points)       $N \leq 50$ .
- Subtask 5 (20 points)      No additional limitations.

# Examples

| input                          | output             |
|--------------------------------|--------------------|
| AB<br>CA<br>58                 | 35<br>23           |
| AABCDEF<br>DBGJCIH<br>12345678 | 7752410<br>4593268 |

# Explanation

In the **first sample case** a possible solution is to map 3 to A, 5 to B and 2 to C (we do not care about the remaining digits). This leads to 35 as the first addend and 23 as the second one, their sum is correctly 58. Other valid solutions may exist (but note that assigning 0 to A or C would produce invalid numbers).

In the **second sample case** a mapping that produces the correct results is the following.

| A | B | C | D | E | F | G | H | I | J |
|---|---|---|---|---|---|---|---|---|---|
| 7 | 5 | 2 | 4 | 1 | 0 | 9 | 8 | 6 | 3 |